

SecurePulse Mobile — Master System Architecture

Mindmap

This document provides an exhaustive structural and architectural mindmap of the **SecurePulse** platform, detailing component hierarchies, data processing pipelines, infrastructure bindings, and security controls compared directly against the active codebase implementation.

Master Architectural Mindmap

flowchart TB

S["SECUREPULSE"]

S --> M[" MOBILE CLIENT"]
S --> B[" CLOUD BACKEND"]
S --> SS[" SECURITY SOURCES"]
S --> P[" PROCESSING PIPELINE"]
S --> AI[" ARTIFICIAL INTELLIGENCE"]
S --> I[" INFRASTRUCTURE"]
S --> SEC[" SECURITY"]

M --> D["Dashboard"]
M --> R["Repositories"]
M --> A["Alerts"]
M --> C["AI Copilot"]
M --> ST["Settings"]

B --> F["FastAPI"]
B --> API["REST API"]
B --> WS["WebSocket"]
B --> DB["PostgreSQL"]

SS --> W["Wazuh"]
SS --> G["GitHub"]
SS --> SEM["Semgrep"]

W --> EV["Security Events"]
G --> EV
SEM --> EV

EV --> P
P --> FIND["Findings"]
P --> ALERT["Alerts"]
P --> SCAN["Scans"]
P --> RISK["Risk & Posture"]

F --> API
F --> WS
F --> DB

WS --> A
WS --> D

API --> D
API --> R
API --> A
API --> C

AI --> C
AI --> LLM["Cloud LLM"]
AI --> OFF["Offline Security Analysis"]

I --> REN["Render Cloud"]
I --> PG["Managed PostgreSQL"]

```
I --> HTTPS["HTTPS / WSS"]

SEC --> JWT["JWT Authentication"]
SEC --> RBAC["RBAC Authorization"]
SEC --> SECRET["Secret Management"]
SEC --> WEB["Webhook Validation"]
SEC --> AUDIT["Audit Logging"]

REN --> F
PG --> DB
HTTPS --> WS

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classDef mobile fill:#111827,stroke:#3b82f6,color:#ffffff;
classDef backend fill:#111827,stroke:#22c55e,color:#ffffff;
classDef security fill:#111827,stroke:#ef4444,color:#ffffff;
classDef ai fill:#111827,stroke:#a855f7,color:#ffffff;

class S main;
class M,D,R,A,C,ST mobile;
class B,F,API,WS,DB backend;
class SEC,JWT,RBAC,SECRET,WEB,AUDIT security;
class AI,LLM,OFF ai;
```



Comprehensive Component Hierarchy Diagram

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graph TD
    subgraph "PRESENTATION TIER (Flutter 3.x Mobile)"
        DASH["Executive Dashboard<br>• Posture Gauge (88-94%)<br>• fl_chart Donut Graph<br>• Se
        REPOS["Monitored Repositories<br>• Health Grades A-F<br>• Branch Tracking<br>• Trigger
        ALERTS["SOC Alert Triage<br>• Severity Filtering<br>• Syslog Inspection<br>• Quarantine
        AI["AI Security Copilot<br>• Streaming Chat UI<br>• CVE-2024-3094 Patches<br>• SQLi / S
        REPORTS["Compliance PDF Engine<br>• SOC 2 / ISO 27001<br>• Time-Stamped Audit ID<br>• P
        SETT["Settings & Diagnostics<br>• Environment Switcher<br>• Server Ping Tester<br>• Bio
    end

    subgraph "STATE & DOMAIN TIER (Riverpod 2.5.1)"
        AUTH_P["AuthStateNotifier<br>• JWT Bearer Token<br>• Biometric Session Gate"]
        DASH_P["DashboardStreamProvider<br>• Telemetry Polling<br>• Cached State Hydration"]
        ALERT_P["WebSocketAlertProvider<br>• Full-Duplex Threat Stream<br>• Event Filtering Sta
        REPO_P["RepositoryScanProvider<br>• Scan Job Status<br>• Findings Aggregator"]
    end

    subgraph "DATA & TRANSPORT TIER"
        API_C["ApiClient (Dio 5.4.1)<br>• 12s Timeout Gates<br>• ApiException Mapping<br>• Dyna
        WS_C["WebSocketService<br>• http->ws / https->wss<br>• Token Handshake<br>• fixed 15-se
        SEC_S["SecureStorageService<br>• Android Keystore (AES-256)<br>• Apple Keychain"]
        HIVE_S["HiveStorageService<br>• securepulse_cache Box<br>• Zero-Latency Offline Read"]
    end

    subgraph "CLOUD BACKEND TIER (Render Cloud)"
        FASTAPI["FastAPI ASGI Server (Python 3.11)<br>• Pydantic Schemas<br>• CORS Whitelisting
        POSTGRES[("Cloud PostgreSQL 16<br>• SSL Mode Require<br>• Connection Pooling")]
        LLM_PROXY["Cloud LLM Gateway<br>• Live AI Prompt Proxy"]
    end

    subgraph "EXTERNAL SECURITY SOURCES"
        WAZUH["Wazuh SIEM Connector<br>• HIDS Syslog Stream<br>• Brute-Force Alarms"]
        GITHUB["GitHub Telemetry API<br>• Codebase Metadata<br>• HMAC-SHA256 Webhooks"]
        SEMGREP["Semgrep SAST Engine<br>• CWE Code Findings<br>• Vulnerability Rules"]
    end

    %% Bindings
    DASH & REPOS & ALERTS & AI & REPORTS & SETT --> AUTH_P & DASH_P & ALERT_P & REPO_P
    AUTH_P & DASH_P & ALERT_P & REPO_P --> API_C & WS_C & SEC_S & HIVE_S
    API_C <--> |HTTPS / REST| FASTAPI
    WS_C <--> |WSS / WebSockets| FASTAPI
    FASTAPI --> POSTGRES
    FASTAPI --> LLM_PROXY
    FASTAPI <--> WAZUH & GITHUB & SEMGREP

```



The following table correlates every mindmap branch with the verified status in the codebase:

Mindmap Branch	Specific Component	Codebase Implementation Reference	Status
Mobile	Executive Dashboard	`lib/features/dashboard/presentation/dashboard_screen.dart`	 IMPLEMENTED
Mobile	Monitored Repositories	`lib/features/repositories/presentation/repositories_screen.dart`	 IMPLEMENTED
Mobile	SOC Alert Triage	`lib/features/alerts/presentation/alerts_screen.dart`	 IMPLEMENTED
Mobile	AI Security Copilot	`lib/features/ai/presentation/ai_assistant_screen.dart`	 IMPLEMENTED
Mobile	Compliance PDF Generator	`lib/core/services/report_pdf_service.dart`	 IMPLEMENTED
Mobile	Settings & Diagnostics	`lib/features/settings/presentation/settings_screen.dart`	 IMPLEMENTED
Cloud	FastAPI Asynchronous Server	Cloud Microservice (`secureguard-backend-7eqm.onrender.com`)	 IMPLEMENTED
Cloud	REST Endpoints Contract	`lib/core/network/api_endpoints.dart`	 IMPLEMENTED
Cloud	WebSocket Threat Engine	`lib/core/network/websocket_service.dart`	 IMPLEMENTED
Cloud	Database Persistence	Cloud PostgreSQL + `lib/core/storage/hive_storage_service.dart`	 IMPLEMENTED
Sources	Wazuh Syslog Ingestion	`lib/features/alerts/data/alerts_repository.dart`	 IMPLEMENTED
Sources	Wazuh Manager REST API	Direct remote daemon control	 PLANNED
Sources	GitHub Telemetry & Webhooks	`lib/features/repositories/data/repository_repository.dart`	 IMPLEMENTED
Sources	GitHub OAuth2 Deep-Links	Interactive in-app OAuth2 callback redirect	 PLANNED
Sources	Semgrep SAST Scanning	`lib/repositories/scan_repository.dart`	 IMPLEMENTED
Sources	Custom In-App Rule Editor	Mobile YAML rule authoring editor	 PLANNED

Mindmap Branch	Specific Component	Codebase Implementation Reference	Status
Processing	Syslog Event Normalization	`lib/features/alerts/models/alert_model.dart`	 IMPLEMENTED
Processing	Alert Lifecycle State Machine	`updateAlertStatus()` in `AlertsRepository`	 IMPLEMENTED
AI	Heuristic CVE Remediation	`AiRepositoryImpl._generateMockResponse()`	 IMPLEMENTED
AI	Cloud LLM Chat Bridge	`POST /v1/ai/chat` client dispatch	 IMPLEMENTED
AI	Autonomous PR Remediation	One-tap GitHub Pull Request patch generation	 PLANNED
AI	On-Device Quantized SLM	Gemma-2B / Llama-3-1B via ONNX Runtime	 PLANNED
Security	Hardware Biometric Gate	`lib/core/services/biometric_service.dart` (`local_auth`)	 IMPLEMENTED
Security	Hardware Keystore Encryption	`lib/core/storage/secure_storage_service.dart` (AES-256)	 IMPLEMENTED
Security	SHA256 PDF Audit Stamping	`ReportPdfService._generateAuditSignature()`	 IMPLEMENTED
Security	Multi-Tenancy per Org	Dynamic organization switching & scoped tokens	 PLANNED
Security	Per-Device Mutual TLS (mTLS)	Client X.509 certificate enrollment	 PLANNED